

Equilibrium of a Rigid Body

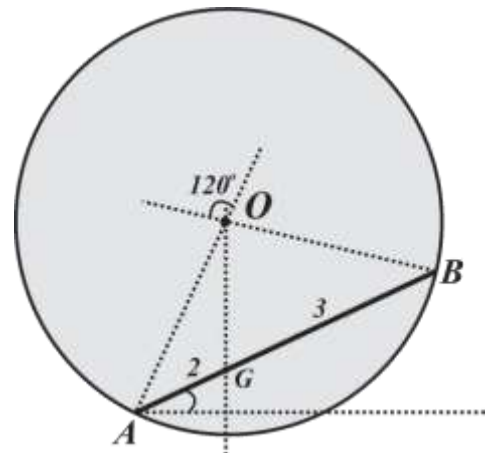
Part A

1. A uniform rod AB of length $2a$ and weight w can turn freely about the end A . It is kept in equilibrium making an angle 30° to the downward vertical by means of a smooth peg at C , where $BC = \frac{3a}{2}$. Show that the magnitude of the reaction on the peg is w .
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2. A uniform rod AB whose length is l is kept in equilibrium so that the rod inclined at an angle 60° to the vertical, with the end A resting against a smooth vertical wall. The rod is being supported by a string attached to the point C on the rod so that $BC = \frac{l}{4}$. The other end of the string is attached to a ring which is fixed to a point vertically above A on the wall. Find the position of the ring. Find also the inclination of the string and the tension in the string.
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3. A uniform rod of weight w , can freely rotate about one end which is jointed to a vertical wall. Other end of the rod is connected by a light inextensible string and kept in equilibrium so that, both the rod and the string inclined 60° to the vertical. Show that the magnitude of the reaction at A is $\frac{\sqrt{3}w}{2}$.
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4. The center of gravity G of a rod AB divide it in the ratio $2:3$. It is kept in equilibrium inside a fixed smooth hollow sphere. (see the diagram) Indicate all the forces acting on the rod. If the angle subtend by the rod AB at O is 120° , show that the inclination of the rod to the horizontal is $\tan^{-1}(\frac{\sqrt{3}}{5})$.
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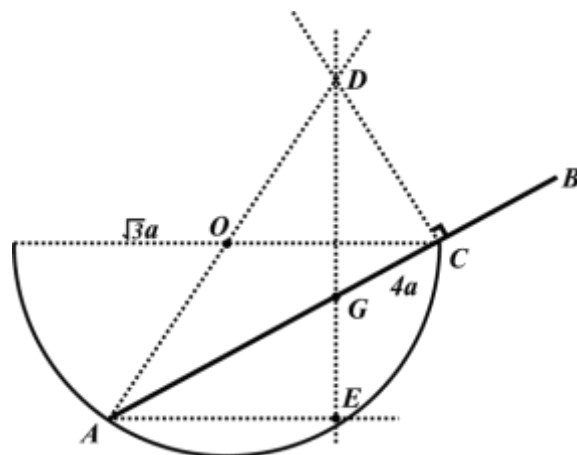


5. A heavy rod is kept inside a fixed smooth hemispherical shell, whose radius is equal to the length of the rod. Center of gravity of the rod divides it in the ratio $2:1$. Show that the inclination of the rod to the vertical is $\tan^{-1} 3\sqrt{3}$.
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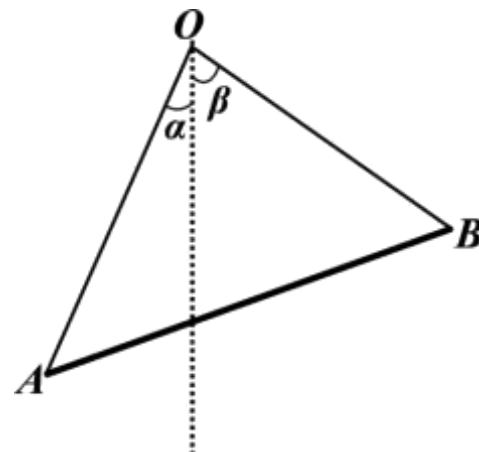
6. One end of a string attached to a point C on a uniform rod AB of weight w and length $8a$, where $AC < 4a$. Other end of the string is fixed to a point D on the wall. The rod is kept in equilibrium by touching the end A on the wall directly below D so that $AD = \sqrt{3}a$. Suppose the inclination of the string to the vertical is 30° , show that the inclination of the rod to the horizontal is $\cos^{-1}(\frac{1}{4})$. Show also that, the tension in the string is $\frac{2}{\sqrt{3}}w$.
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7. Following diagram shows the cross section of a uniform rod AB of length $4a$ which is in equilibrium inside a fixed hollow hemisphere of radius $\sqrt{3}a$, whose rim is horizontal. The point C on the rod is in contact with the rim of the hemispherical shell. Show that the inclination of the rod to the horizontal is $(\frac{\pi}{6})$.

Show also that, the length of CB is one fourth of total length of the rod AB .



8. A rod AB of weight w is kept hanging by means of a light inextensible string of AOB , which is passing over a smooth peg at O as show in the diagram. Show that in the position of equilibrium, both the pieces of strings make same angle with the vertical. If the center of gravity of the rod divides it in the ratio $2: 1$, show further that, the ratio between the lengths of the two pieces of string is also $2: 1$.

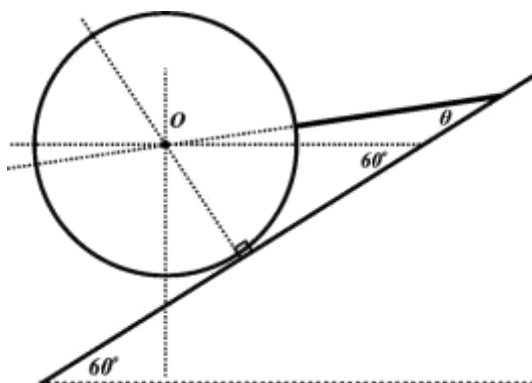


9. A uniform rod can freely rotate about its upper end. The other end is attached to a string fixed to a point, located as same level as the upper end of the rod. The length of the rod is equal to the distance between the upper end and the fixed point. If, the tension in the string is equal to the weight of the rod, show that the inclination of the rod to the horizontal is $\tan^{-1} (\frac{1}{2})$.

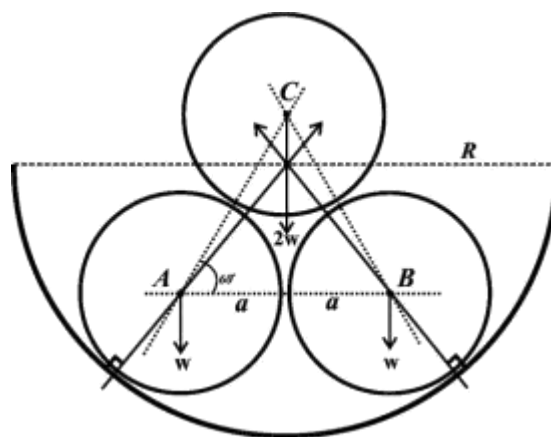
10. The center of gravity G of a rod AB of length a is such that, $AG = \frac{a}{4}$. It is in equilibrium at an angle θ to the horizontal with its two end touching two smooth planes each inclined at an angle α to the horizontal. The line of intersection of the two planes is perpendicular to the rod. Prove that, $\cot \alpha = 2 \tan \theta$.

11. A rod of length $5l$ is suspended by means of a string of length $7l$ which is attached to the ends of the rod; the string being passing over a smooth peg at P . If the center of gravity divides the rod in the ratio $4: 3$, show that two strings at P are perpendicular to each other.

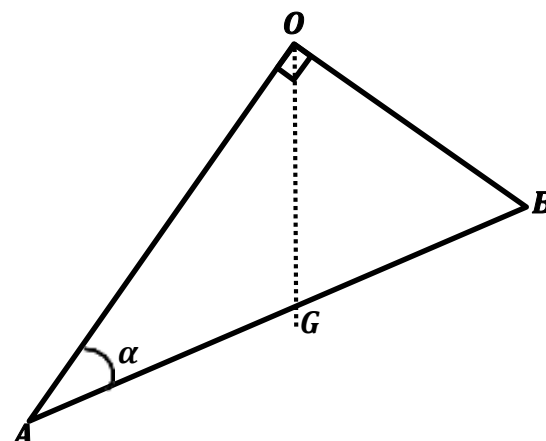
12. A uniform sphere of rests on a fixed smooth plane inclined at an angle 60° to the horizontal, being supported by a light inextensible string of whose length equal to the radius of the sphere as shown in the diagram. one end of the string is attached the point on the surface of the sphere and the other end attached to a point on the plane. Show that the angle θ made by the string with the plane is 30° . Show also that, tension in the string is equal to the weight of the sphere.



13. Three spheres A , B and C each having radius a and weight w , w and $2w$ respectively are kept symmetrically inside a smooth hemispherical vessel of radius R , so that sphere C is on A and B as shown in the following diagram. For equilibrium show that, $(\sqrt{13} + 1)a > R$.



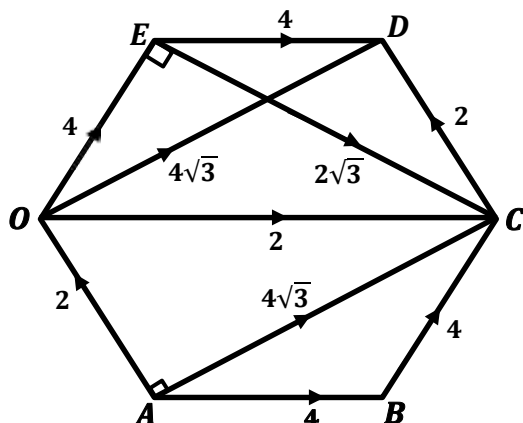
14. A uniform rod AB of length $2a$ and weight W , suspended from a fixed point O by two light inextensible strings OA and OB , is in equilibrium as shown in the figure. G is the mid-point of AB . It is given that, $\angle AOB = \frac{\pi}{2}$ and $AB = a$. Show that $\angle AOG = \alpha$, and find the tensions in the two strings. (2017 A/L)



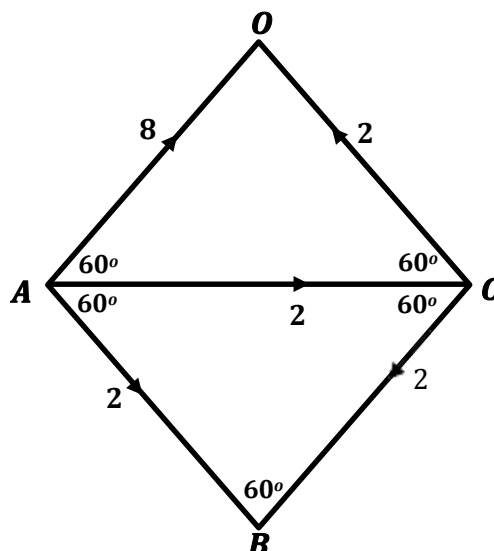
Part B

1. Explain and define the term “Rigid body” in mathematics.
2. The system of forces measured in Newton’s (N) , act on a body as show in the following diagrams. By considering the horizontal and vertical components of the forces find the resultant and the direction of the resultant in each case. Calculate also the moment of the system of forces about O and A .

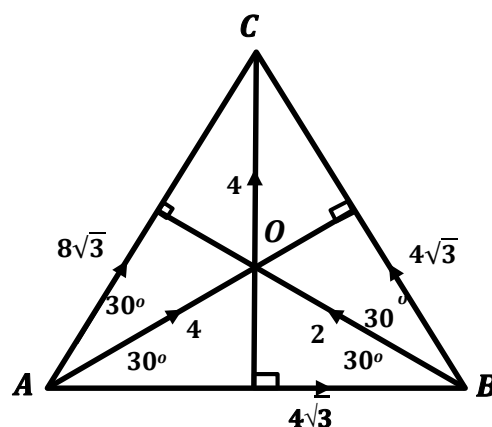
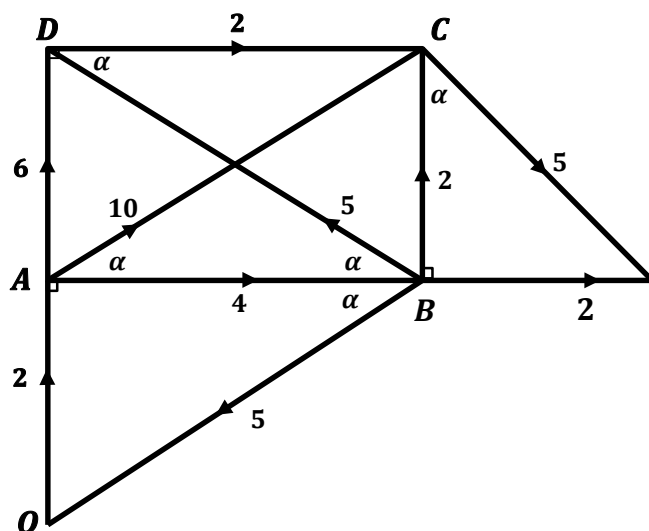
a $ABCDE$ is a regular hexagon with $AB = 2\text{m}$.



b ABC is a triangle with $AB = 2\text{m}$.



c $ABCD$ is a rectangle with $AB = 4\text{m}$ and $BC = 3\text{m}$. **d** ABC is a triangle with $AB = 1\text{m}$.



3. Obtain the resultant of two parallel forces and its point of action, when they are homogeneous and heterogeneous forces.
4. Explain the conditions satisfying by the forces , when a rigid body is in equilibrium under two forces. Extend the knowledge when, a rigid body is in equilibrium under three forces for the cases they are non-parallel and parallel.
5. A uniform rod AB of length $6a$ and weight w kept horizontally by means of two smooth pegs P and Q . If, $AP = 2a$ and $AQ = 5a$, show that the ratio between the magnitude of the normal reaction on P and Q is 2: 1.

6. A uniform rod AB of weight 12 kg and length 8 m kept equilibrium by end A touching a smooth vertical wall. The rod is supported by a light string attached to the point C on the rod and the other end of the string being attached to a point D vertically above A so that $AD = 2\text{ m}$. If $\angle DAB = 60^\circ$, show that, the inclination of the string to the vertical is also 60° . Using the Lami's theorem, show that the tension in the string is of magnitude $24g\text{ N}$ and the reaction between the wall and the rod is of magnitude $12\sqrt{3}g\text{ N}$.
7. A smooth peg C is fixed at a horizontal distance $\sqrt{3}a$ from a smooth vertical wall. A uniform rod of weight w and length $\frac{16}{3}a$ is kept in equilibrium against the wall and the peg in a vertical plane through the center of the gravity of the rod. Show that the inclination of the rod to the horizontal is 30° . Show also that the magnitude of the reaction at the peg is $\frac{2w}{\sqrt{3}}$.
8. The end A of a uniform rod AB of weight W touches a vertical wall. One end of a string is tied to the rod at a point C where $AC = \frac{1}{4}AB$. The other end of the string is tied to a point on the wall vertically above A . If the rod rests in a position inclined to the vertical, find the length of the string in terms of AB . (applying the *sin rule* for suitable triangles).
9. A smooth peg is fixed at a point P at a distance a from a smooth vertical wall. A uniform rod AB of length $6a$ and weight W is in equilibrium resting on the peg with the end A in contact with the wall. Taking θ to be the angle made by the rod AB with the horizontal, find the reaction at P , in terms of W and θ . Show that, $3 \cos^3 \theta = 1$.
10. AB is a rod of weight W . Its end A touches a smooth vertical wall. One end of a light inextensible string is joined to the end B of the rod and the string is joined to a point C on the wall located vertically above A . In the position of equilibrium the rod makes an angle 60° with the downward vertical. If the center of gravity of the rod divides it in the ratio $2:1$ from end A , show by using *cot - rule* for a suitable triangle that, the inclination of the string to the vertical is 30° . Using Lami's theorem, show that the tension in the string and the reaction between the wall and the rod is of magnitude $\frac{2}{\sqrt{3}}W$ and $\frac{1}{\sqrt{3}}W$ respectively.
11. A rod AB whose center of gravity at G with $AG:GB = a:b$ ($b > a$) rests in a vertical plane inside a fixed smooth hollow spherical shell. The rod is inclined at an angle θ to the horizontal and the rod subtends an angle 2α at the center of the sphere, show by applying *cot -rule* for a suitable triangle that, $\tan \theta = \left(\frac{b-a}{b+a}\right) \tan \alpha$.
12. The end A of a uniform rod AB of weight W is hinged to a fixed point. The rod is held in equilibrium in a horizontal position by means of a string attached to B and to a point C vertically above A . The string is inclined at 60° to the rod. Find the tension in the string and the reaction of the hinge.
13. A uniform rod AB of weight w is smoothly hinged at A and kept in equilibrium making an angle θ to the vertical, by applying a horizontal force of magnitude $\frac{w}{3}$. Show that, $\tan \theta = \frac{2}{3}$ and the magnitude of the reaction of the hinge at A is $\frac{\sqrt{10}}{3}w$.

14. A uniform rod AB with the end A touching a smooth vertical wall is held in equilibrium in a vertical plane perpendicular to the wall by means of a light inextensible string passing through a fixed smooth ring C vertically above A , with one end of it joined to the rod at A and the other end at B . Let $BAC = \pi - \alpha$ and $ACB = \beta$. Sketch the cross section, showing the position of equilibrium of the rod. Find the tension in the string, by considering the vertical forces and by taking the moment about A for equilibrium of the rod. Hence find a relation between α and β .
15. The end A of a uniform rod AB weighing W and of length a is hinged freely to a vertical wall. The end B of the rod is connected to point C on the wall by means of a light inextensible string of length a . C is at a distance b vertically above A . Show that the reaction at of the hinge at A is $\frac{W}{4b} \sqrt{a^2 + 8b^2}$. Find also the tension in the string.
15. A uniform rod AB of length $8a$ and weight w can turn freely about the end A which is joined to a vertical wall. The rod is held in a horizontal position by means of a rope of length $5a$, with one end attached to a point D on the rod placed distance $3a$ away from A and the other end of the string attached to a point C on the wall at directly above A . Show that the tension in the string is $\frac{5w}{3}$ and magnitude of the reaction at A is $\frac{\sqrt{10}w}{3}$.
16. AB is a uniform rod of weight W which can turn freely about a hinge at A . It is held in equilibrium by means of a weight P attached to a string BCP passing over a smooth peg C vertically above A . If $AC = AB$ and $ACB = \theta$, show that $P = W \cos \theta$. Find the magnitude and the direction of the reaction at the hinge.
17. One end of a rod whose weight is W , smoothly hinged to a point on a vertical wall. It is borne by a string attached to the other end of the rod which connects a point at same horizontal level as the hinge. The rod and the string are both inclined at the same angle α to the vertical. Show that the resultant action at the hinge is $\frac{W}{4} \sqrt{9 + \tan^2 \alpha}$.
18. The upper end A of a uniform rod AB which is in equilibrium inclined at an angle α to the horizontal touches a smooth peg. One end of a string attached to a point C , on the same level as A and the other end is attached to the lower end B of the rod. If β is the inclination of the string to the horizontal, prove that, $\tan \beta = 2 \tan \alpha + \cot \alpha$.
19. A uniform rod of length a hangs one end against a smooth vertical wall and the other end being supported by a string of length l which is fixed to a point on the wall. Show by applying cot rule for a suitable triangle that, the rod can rest inclined to the wall at an angle θ ($< \frac{\pi}{2}$) given by $\cos^2 \theta = \frac{l^2 - a^2}{3a^2}$. Deduce that, $\frac{1}{2} < \frac{a}{l} < 1$ for equilibrium.
20. A heavy non-uniform rod of length $3a$ is kept in equilibrium against a smooth vertical wall. Other end of the rod is supported by a string of length $6a$. If the position of the center of gravity of the rod is 2: 1, show that the angle between the rod and the wall is $\sin^{-1} \left(\frac{\sqrt{5}}{2\sqrt{2}} \right)$.

21. A rod whose center of gravity divides it into two portions whose lengths are a and b , has a string of length l tied to its two ends and the string is slung over a smooth peg. Find the inclination of the pieces of strings, of the rod at the position of equilibrium, in which it is not vertical.
22. A beam AB of weight w is divided by its center of gravity G into two portions AG and GB whose lengths are a and b respectively. The beam rests in a vertical plane through G , on a smooth floor AD and against a smooth vertical wall DB . Ends of a string are attached to a hook at D and to the beam at a point P . Let θ and ϕ be the inclinations of the beam and the string to the horizontal respectively. Find the normal reaction R at the end B on the rod in terms of the tension T in the string, by considering the horizontal component of forces of the rod for equilibrium. Find R using other method and show that, $T = \frac{wa \cos \theta}{(a+b) \sin(\theta-\phi)}$.
23. Two equal smooth spheres each of weight W and radius a are in equilibrium inside a smooth spherical vessel of radius $3a$. Show that the reaction between the vessel and a sphere is twice the reaction between the two spheres.
24. A sphere of weight W rests between two smooth planes. One plane is vertical and the other is inclined at an angle α to the vertical. Find the reaction of the planes.
25. Two spheres each of weight W and radius a are joined by a string of length $2l$ at two points on the surface of each sphere. The two spheres are in equilibrium symmetrically touching each other and with the string passing over a smooth peg. Show that the tension in the string is $\frac{W(a+l)}{\sqrt{l^2+2al}}$ and that the reaction between the spheres is $\frac{aW}{\sqrt{l^2+2al}}$.
26. Two smooth spheres of radius r and weight W are kept inside a smooth hollow cylinder of diameter $3r$ and whose axis vertical. Sketch the diagram of a cross section for equilibrium of two spheres, indicating all the forces acting on spheres. Find forces exerted on the wall and the bottom of the cylinder due to the spheres.
27. The curved surface of a solid hemispherical bowl of radius r and weight w is kept in equilibrium on a smooth horizontal table by means of a light inextensible string whose length is l ($< r$), one end of which is attached to the point on the rim of the sphere and the other end to a point on the table, so that its face is inclined at an angle θ with the horizontal. Show that the string should be vertical for the body to be in equilibrium. Find the value of θ and show that the tension in the string is given by $\frac{3}{8} w \frac{r-l}{\sqrt{2lr-l^2}}$.
28. A hemispherical bowl of weight W is kept on a smooth horizontal table. A particle of weight $\frac{W}{\sqrt{3}}$ is hanged at a point on a rim of the bowl. Find the angle that the circular base of the bowl makes with the horizontal in equilibrium position.
29. A rod AB of length $3a$ is placed inclined at an angle α to the horizontal, inside a sphere in a vertical plane through its center. The rod subtends an angle β at the center of the sphere. If the center of gravity of the rod lies at a distance a from A show that, $\tan \frac{\beta}{2} = 3 \tan \alpha$.

30. A heavy uniform rod of length $2a$ rests partly within a fixed smooth hemispherical bowl of radius r . The rim of the bowl is horizontal and one point of the rod is in contact with a point of the rim. If θ be the inclination of the rod to the horizontal, show that, $2r \cos 2\theta = a \cos \theta$. Deduce that, $2r > a > \frac{2r}{\sqrt{6}}$.
31. Thin hemispherical bowl of radius a fixed with its rim uppermost and horizontal. A smooth uniform rod AB of weight W and length $2l$ ($> 2a$) rests with the end A of the inner surface of the bowl and a point C of the rod in contact with the rim. Indicate the forces acting on the rod. By taking moments about A , show that the reaction R at C is of magnitude $\frac{Wl}{2a}$. Also, obtain another relation between R and W . Hence, show that the length CB is $(\frac{7l - \sqrt{l^2 + 32a^2}}{4})$.
32. A uniform solid hemispherical bowl of radius a and weight w is suspended by means of a light inextensible string one end attached to a point of a smooth vertical wall and other end attached to a point of an edge of the circular base of the bowl. The bowl is in equilibrium so that its curved surface against the wall. Suppose the inclination of the plane surface is α to the vertical show by applying *sin rule* for a suitable triangle that, $\tan \alpha = \frac{3}{8} + \tan \theta$, where θ is the angle in which the string inclined to the vertical.
33. A uniform rod of weight W and length $4a$ rests on a vertical plane inside a smooth spherical shell of radius $2\sqrt{2}a$. A particle of weight w is now attached to a point on the rod at distance a away from the center of gravity of the rod. If the inclination of the rod to the horizontal is θ , in the position of equilibrium, show that $\tan \theta = \frac{w}{2(W+w)}$.
34. A uniform rod of weight W and length $2l$ rests in a vertical plane inside a smooth hollow sphere of radius r ; ($r > l$). A particle of weight w_0 is attached to a point on the rod, which is located at a distance d away from the center of gravity of the rod. Find the position of the resultant of W and w_0 intersects with the rod. If θ is the inclination of the rod to the horizontal, show by using a *cot rule* for a suitable triangle that, $\tan \theta = \frac{w_0 d}{(W+w_0)\sqrt{r^2 - l^2}}$.
35. A fixed smooth hemispherical bowl of diameter a is placed so that its edge touches a smooth vertical wall. A heavy uniform rod is in equilibrium inclined at 60° to the horizontal with one end resting on the inner surface of the bowl and the other end resting against the wall. Show that the length of the rod is $a + \frac{a}{\sqrt{13}}$.
36. A smooth fixed cylinder of base radius r and center O lies parallel to a smooth vertical wall at a distance c from O with its axis horizontal. A heavy uniform rod AB of length $2a$ is kept on the cylinder in a vertical plane making a right angle with the axis of the cylinder. In the position of equilibrium, end A of the rod touches the wall and end B located higher level than A . If the rod makes an angle θ with the vertical, show that, $c = a \sin^3 \theta + r \cos \theta$.
37. The center of gravity of a heavy rod divides it in two portions of length a and b . The rod rests in a vertical plane, inside a hollow hemispherical shell of radius r . Show that the inclination of the rod to the horizontal is $\tan^{-1}(\frac{b-a}{\sqrt{4r^2 - (a+b)^2}})$.

38. A rod of weight W and length b is kept inside a fixed spherical shell of radius a . Sketch a cross section of the system, indicating all the forces acting on it when it is in equilibrium in a vertical plane through the center of the sphere. Find the magnitudes of the reaction at each end of the rod. Show that the rod is divided by its center of gravity in the ratio, $\left(\frac{b-c}{b+c}\right)$; where $c = \sqrt{4a^2 - b^2} \tan \theta$, where θ is the inclination of the rod to the horizontal.
39. A rod of length $a + b$ rests wholly within a smooth hemispherical bowl of radius r ($> \sqrt{ab}$) and center O . Its center of gravity G divides the rod in to two portions of length a and b ($> a$). If θ be the inclination of the rod to the horizontal at the position of equilibrium, show that, $\sin \theta = \frac{b-a}{2\sqrt{r^2-ab}}$. Show also that, $OG = \sqrt{r^2 - ab}$.
40. The extreme diameter of a uniform semicircular plate is AB . This plate is suspended from A by means of a inextensible and light string. Find the angle made by AB with the vertical.
41. A solid cone of base radius r and height h is in equilibrium being suspended from a point on the edge of its base. Find the angle made by the axis of the cone with the vertical.
42. When a uniform hollow hemisphere is in equilibrium when suspended from a point on its edge, show that the angle made by the plane containing the edge with the vertical is $\tan^{-1} \left(\frac{1}{2}\right)$.
43. Both ends of a light inextensible string are attached to two vertices of an equilateral triangular lamina whose length of a side is $2a$. Lamina is suspended by means of the string against a smooth peg and kept in equilibrium such that a side of the lamina is vertical. Show that the inclination of the each pieces of strings makes an angle 30° with the vertical. Show also that the length of the string is $2\sqrt{3}a$. Also find the tension in the string.
44. Three uniform smooth cylinders of radius r and weight w are kept symmetrically inside a fixed smooth cylinder of radius R , such that one cylinder is on the other two cylinders. Suppose axes of all the cylinders are horizontal show that, the equilibrium is possible only if, $r \geq \frac{R}{2\sqrt{7}+1}$.
45. Two planes inclined at an angle α with the horizontal met at horizontal line l forming a smooth basin. A , B and C are three smooth cylinders each of weight w . The cylinders A and B kept inside the basin so that their axes parallel to l . Each cylinders are in equilibrium with touching each other and the planes. Then the cylinder C kept on A and B symmetrically with its axis parallel to l . Show that, the reaction between A and B is $w \left(\frac{3}{2} \tan \alpha - \frac{1}{2\sqrt{3}} \right)$. Show also that equilibrium is possible only if, $\tan \alpha > \frac{\sqrt{3}}{9}$.
46. A heavy rod of weight of w is kept in equilibrium with it's two ends touching two smooth planes inclined at an angle α and β to the horizontal. The line of intersection of the two planes is perpendicular to the rod. If the center of gravity of the rod divide it in the length a and b find the inclination of the rod to the horizontal. Show also that the magnitude of the reaction between the rod and the planes are given by $\frac{w \sin \alpha}{\sin(\alpha+\beta)}$ and $\frac{w \sin \beta}{\sin(\alpha+\beta)}$.

47. A smooth uniform solid sphere of radius a and weight w is hanged at O by means of a light inextensible string of length a . One end of a uniform rod weight w and length $4a$, is also connected to the point O . The system is kept in equilibrium so that rod and the sphere touch each other. Show that, both rod and the string make same angle to the vertical and it is equal to $\frac{\pi}{12}$. Show also that the tension in the string can be expressed as $\frac{w \cos(\frac{\pi}{12})}{\sin(\frac{\pi}{3})}$ and the normal reaction between the rod and the sphere is $\frac{w \sin(\frac{\pi}{12})}{\sin(\frac{\pi}{3})}$.
48. Two identical uniform ladders AC and CB are smoothly jointed together at C and kept on smooth floor by touching the ends A and B . The ladders are supported by a light inextensible string which connects the mid points of the rods. A man, whose weight is similar to one of the ladder, starts to climb up on AB . Show that, the greatest reaction out of reactions A and B is exerted at, in any position of the man on the ladder.
49. Smooth uniform rod AB of length $2a$ and weight w can freely rotate about the end A . A ring C of weight $2w$ can freely slide on the rod. The rod is kept in equilibrium by attaching a string of length $\frac{a}{4}$ one end to the ring C and the other end to the point D found on same level as A . Suppose $AD = \frac{a}{4}$ find the reaction between the rod and the ring. Show that the inclination of the rod to the horizontal is $\frac{\pi}{3}$.
50. A light smooth ring Q is free to slide on a uniform smooth rod AB of weight w and length $2a$. The rod is hanged and kept in equilibrium by means of a light in extensible string AOQ of length $2l$, passing over a smooth peg at O , with one end connected to the end A and the other end being attached to the ring. Suppose the inclination of the rod to the horizontal is θ , show by considering the equilibrium of the ring and the rod that,
- each part of the string is inclined at an angle θ to the vertical.
 - tension in the string is $\frac{w \sec \theta}{2}$.
 - $a \cos^3 \theta = l \sin \theta$.
51. AB is a light wire of the form of a circular arc, which subtends an angle $\frac{\pi}{3}$ at the center O . Two particles of weight $\frac{3\sqrt{3}w}{2}$ and $\sqrt{3}w$ are suspended to the ends A and B respectively and kept on a smooth horizontal plane in equilibrium in a vertical plane through the center O . Show that the angle made by OB to the vertical is $\tan^{-1} \left(\frac{\sqrt{3}}{4} \right)$.
52. A smooth uniform hemispherical bowl of radius a is fixed rigidly with its axis horizontal. A particle of weight W is connected to the end A of a uniform rod AB of weight $2W$ and of length l and kept in equilibrium, in a vertical plane through the center of the bowl, with the end B touching a point inside the bowl and with the other end A extending outside the edge. If the rod is inclined at angle θ to the horizontal show that, $l = 3a \cos 2\theta \sec \theta$.

53. If three non-parallel coplanar forces acting upon a rigid body keep it in equilibrium, show that they must meet in a point. Two uniform smooth spheres of different radii with centers A and B and having equal weight W rests inside a fixed smooth right circular hollow cone which has its vertex downwards and each sphere touches the cone at one point only. The semi vertical angle of the cone is $\frac{\pi}{3}$ and its axis makes an angle β ($< \frac{\pi}{6}$) with the vertical. If the line AB makes an angle θ with the upward vertical, show that, $\tan \theta = (\cot 2\beta - \frac{1}{2} \operatorname{cosec} 2\beta)$.

54. A smooth hemispherical bowl of radius r , rests on a smooth horizontal table and partly inside it rests a smooth uniform rod, of length $2l$ and of weight equal to that of the bowl. In the position of equilibrium, the inclination of the base of the hemisphere to the horizontal is α ($< \frac{\pi}{2}$) and the angle subtended at the center by the part of the rod with in the bowl is 2β ($> \frac{\pi}{2}$). Show that,

i. $l \sin(\alpha + \beta) = r \sin \alpha$

ii. $\cot \alpha = \tan 2\beta - \frac{1}{2} \sec 2\beta$

55. A uniform sphere of radius a and weight w rests on a fixed smooth plane inclined at an angle α to the horizontal, being supported by a light inextensible string of length l , one end of which is attached the point on the surface of the sphere and the other end to a point on the plane. Find the angle θ made by the string with the plane. Construct a triangle of forces for those acting on the sphere. Using this triangle of forces or otherwise, show that,

i. the tension in the string is $w \frac{(l+a) \sin \alpha}{\sqrt{l^2 + 2al}}$ and

ii. the reaction from the plane is $\frac{w \cos(\alpha - \theta)}{\cos \theta}$.

56. One end of a light inextensible string whose length is l is attached to the vertex of a right uniform solid circular cone of radius r and height h and the other end of the string is attached to a point on the circumference of the circular base of the cone. Cone is suspended and held in equilibrium by means of the string against two smooth pegs fixed on the same horizontal levels, length d apart, where $h > d$, such that, the axis of the cone is horizontal.

Show that, $l = d + \frac{(h-d)\sqrt{h^2 + 4r^2}}{h}$.

57. A right solid cone of height h and semi vertical angle is kept in equilibrium by touching its base on a horizontal wall. Its supported by a string of length attached to a vertex of the cone and the other end of the string being attached to a point on the wall. Show that for equilibrium,

$h \sec \alpha \leq l \leq \frac{h}{3} \sqrt{9 + 16 \tan^2 \alpha}$.

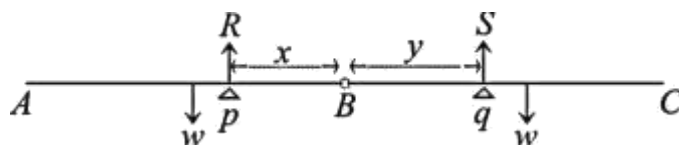
Smoothly Jointed Rods

Part A

1. AB and BC are identical uniform rods each of weight W hinged freely from the point B . The inelastic string connecting the mid points of the rods are so long that when straight angle ABC is a right angle. If the system is freely hanging from the point A , and is in equilibrium show that by taking moments about A for equilibrium of the system, the inclination of the rod AB to the vertical is $\tan^{-1}(\frac{1}{3})$. By taking moments about B for equilibrium of the rod BC show that the tension in the string is of magnitude $\frac{3W}{\sqrt{5}}$

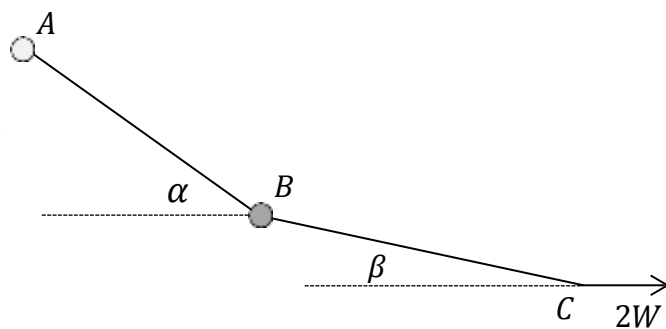
2. Two rods AB , BC each of weight w and length $2a$ are smoothly jointed at end B and kept symmetrically in equilibrium with the rods being supported by two smooth pegs which are located in same horizontal level at $2b$ apart so that joint B is upper most. Show that the inclination of each rod to the horizontal is α is given by, $a \cos^3 \alpha = 2b$.

3. Two rods AB and BC of weight w and length $2a$ are smoothly jointed at B and kept horizontally by means of two pegs p and q on rods AB and BC respectively, as shown by the following figure. If $BP = x$ and $BQ = y$ show that, $a(x + y) = 2xy$. Show also that, $y \geq \frac{2a}{3}$. Find the reaction at p, q and also at joint B in terms of a and x .



Part B

1. Three identical uniform rods are connected smoothly at their ends A, B and C to form a triangular frame work ABC . The frame work is freely subtended at the end A and kept in equilibrium. Find the reaction at either ends B or C . Determine the inclination of the reaction to the horizontal. (length $2a$ and weight W each)
2. Two uniform rods AB and BC each of length $2a$ and weight w are smoothly jointed at B . The ends A and C are smoothly hinged to two points on the same horizontal line at distance $2a$ apart. The system hangs in equilibrium with B below AC . Show that the reaction at the rod AB at the joint B is a horizontal force of magnitude $\frac{\sqrt{3} w}{6}$.
3. Two uniform rods of same length and weights W and $2W$ are joined smoothly at B . Rods AB and BC are inclined an angle α to the horizontal by an inelastic string connecting the mid points of AB and BC . The A and C ends are on a smooth horizontal plane while ABC in a vertical plane. Find the components of the action of B , reaction of A and C . Show that the tension in the string is $\frac{3}{2}W \cot \alpha$. Show also that, the inclination the action of B is $\tan^{-1}(\frac{1}{6}(\tan \alpha))$ to horizontal.
4. Four rods AB, BC, CD and DA of equal lengths are freely jointed so as to form a rhombus. The rods AB and AD are light and the rods BC and CD are each uniform and of weight w . The rhombus is suspended at A and kept in position by another equal light rod BD freely jointed at B and D . Show that the reaction at the joint C is of magnitude $\frac{\sqrt{3} w}{6}$. Show also that the thrust in the rod BD is $\frac{\sqrt{3} w}{2}$.
5. Two uniform rods AB, BC of length $2a$ and weight W are smoothly kept with the ends of A and C on a smooth horizontal plane. ABC is placed vertically by means of an inelastic string which connects A and the mid-point D of BC . The system is in equilibrium with the angle $ABC = 2\alpha$. Show that the reaction at the joint B is of magnitude $\frac{w}{4}\sqrt{1+9\tan^2\alpha}$. Show that the distance $AD = a\sqrt{1+8\sin^2\alpha}$ and deduce that the tension in the string is $\frac{W}{4\cos\alpha}\sqrt{1+8\sin^2\alpha}$.
6. Two equal, uniform rods AB and AC of length $2a$ and weight W are jointed smoothly at A and the two ends B and C are placed on a smooth horizontal table in a vertical plane. The equilibrium of the system is maintained using a light inextensible string joined between point C and the mid-point of AB . If, each rod is inclined at an angle $\alpha (< \frac{\pi}{2})$ to the horizontal, show that the magnitude of the reaction at A is $\frac{W}{4}\sqrt{1+9\cot^2\alpha}$. Find also the tension in the string.
7. Uniform rods AB and BC of length $2a$ and weight W are hinged smoothly at B . A is connected by a stationary smooth pivot at A with rods AB and BC inclined at an acute angles α and β to the horizontal respectively by means of a horizontal force of $2W$ at C as shown by the following diagram. Show that, $\tan \beta = \frac{1}{4}$ and $\tan \alpha = \frac{3}{4}$.
Show also that the reaction at B is $\sqrt{5} W$.



8. OA and AB are uniform rods each of length $2a$ and weight $2W$ and $3W$ respectively are hinged smoothly at A end and pivoted stationary and freely hanging from O . System is kept at equilibrium by a horizontal force of P at B . OA and AB rods are vertically inclined at angles of α and β .
Show that, $P = \frac{3W}{2} \tan \beta$. Find the reaction at A and the hinge at O with the inclination of reactions to horizontal. Show that, $3 \cot \alpha \tan \beta = 8$.
9. Four uniform rods of length $2a$ and weight W are smooth hinged at the points A, B, C and D and hanged freely from A . An inelastic string connects the points A and C so that $ABCD$ forms a square. The system is in equilibrium. Show that the tension in the string is of magnitude $2W$.
10. Four uniform rods AB, BC, CD and DA of length $2a$ and weight w are smoothly jointed at A, B, C and D . The framework is freely suspended at A and the mid-points of AD and DC are connected by a light inextensible string. When the frame work rests in equilibrium, $BAD = 2\alpha$.
Show that the reaction at the joint C is $\frac{w}{2} \sqrt{4 + \tan^2 \alpha}$ and the tension in the string is $4w$.
11. Four uniform rods of length $2a$ and weight W each are smoothly hinged at the points A, B, C and D . A light rod connecting the midpoints of the rods BC and CD is hanging from A so that angle $BAD = 2\alpha$. Find the reactions at the free joints. Show that the thrust in the light rod is $4W \tan \alpha$.
12. Four uniform rods of weight W each are hinged smoothly to form a rhombus $ABCD$. The system is suspended freely from D . Joint A and the mid-point of the rod CD are joined by a light rod and given a shape of a rhombus such that, $ABC = 60^\circ$.
Find the magnitude of the reaction at C and show that the thrust in the light rod is $2W$.
13. Four uniform rods AB, BC, CD and DE each of length $2a$ are smoothly jointed at B, C and D .
The weights of each of the rods AB and DE is $2W$ and the weight of each of the rods BC and CD is W . The rods are suspended in a vertical plane from the points A and E on the same horizontal level and the system is in equilibrium with the rods AB and BC making angles, α and β respectively, with the vertical. Show that, $\tan \beta = 4 \tan \alpha$.
14. Six uniform rods each of length $2a$ and weight W are smoothly jointed at their ends to form the framework $ABCDEF$. The framework is freely suspended from A and given the shape of a regular hexagon by means of two light rods BF and CE .
Find the reaction at the joint D and show that the ratio of the thrusts in the rods BF and CE is $5:1$.
15. Six uniform rods each of length $2a$ and weight W are smoothly jointed at A, B, C, D, E and F . The system is kept in equilibrium as a hexagon by a light horizontal rod PQ with smooth hinges at two points of the rods BC and EF . The whole system is hanging from A . Find the reaction at the free joints.
Show that the thrust in the light rod is $3\sqrt{3}w$. Show that, $BP = \left(\frac{a}{3}\right)$.
16. AB, BC and CD are three uniform rods of equal weight and length, smoothly hinged at B and C . The ends A and D are hinged to fixed smooth horizontal pins at the same level. The system hangs in equilibrium. If AB and CD are inclined at the same angle α to the horizontal, and β is the inclination of the reaction at A on AB to the horizontal, show that, $\tan \alpha = \frac{2}{3} \tan \beta$.

17. Five rods each of unit weight w are smoothly joined to their ends to form the framework, $ABCDE$ where $ED = DC = a$ and $AE = BC = b$. The framework is kept in equilibrium on a horizontal table by means of the side AB and given the shape of a pentagon with, $EAB = ABC = 120^\circ$, and $BCD = AED = 90^\circ$ by attaching a light string to the mid points of the sides AE and BC . Show that the tension in the string is $\frac{(b+5a)w}{\sqrt{3}}$. Suppose the string is attached to EC (instead of the mid points of AE and BC) calculate the tension in the string.
18. AB, BC, CD, DE and EA are five uniform rods each of weight W are jointed smoothly at their ends. They make a convex pentagon by smoothly joining another light rod BE . The frame work is suspended from A and the rods AB and AE make an angle θ with the vertical. The rods BC and DE make an angle φ with the vertical. Show that the vertical and the horizontal components of the reaction at C are $\frac{W}{2}$ and $W \tan \varphi$ respectively. Show also that, the stress in the rod BE is of magnitude $W(2 \tan \theta + \tan \varphi)$.
19. Six uniform rods each of length $2a$ and weight W are smoothly jointed at their ends to form a framework $ABCDEF$. The framework is freely suspended from A and given the shape of a regular hexagon by means of three light rods AC, AE and CE . Show that the reaction at the joint D is horizontal and of magnitude $\left(\frac{W}{2}\right)$. Find also that the tension the rod AC or AE and thrust in the rod CE .
20. Two equal uniform heavy rods AB and BC are jointed smoothly at B . The system is suspended from the point A and the mid points of the rods are connected by a light inextensible string, such that AB and BC are perpendicular to each other. If system is in equilibrium with the rod AB making an angle θ with the downward vertical, show that, $\tan \theta = \frac{1}{3}$. Show also, that the reaction at B is along the direction BC .
21. Four equal uniform rods PQ, QR, RS and SP each of weight W , are smoothly jointed to form a rhombus $PQRS$. It is suspended from the point P and kept in equilibrium in a vertical plane with the angle QRS equal to $2 \tan^{-1}(2)$ by light inextensible string connected to the midpoint of the rods PS and RS .
- Find the reaction at the joints R and S .
 - Show that the tension in the string is $4W$.
22. Four equal, uniform rods of length $2l$ and weight W are jointed smoothly at A, B, C and D , the framework is suspended from A . It is kept in equilibrium forming a symmetrical shape about the axis AC , by means of passing a light inextensible string which forms a loop through light rings fixed at the mid-points of rods AB and AD and through C . If $BAD = 60^\circ$ then find the horizontal and vertical components of the reaction at B . Also find the tension in the string.
23. AB and BC are two rods of equal length smoothly jointed at B . The weight of AB is W and the weight of BC is $2W$. The point A and the midpoint D of BC are connected by a light string and the framework is in equilibrium in a vertical plane through the centers of gravity of rods with $ABC = 60^\circ$ and the ends A and C on a smooth horizontal plane. Find the reactions at A and C , the tension in the string, and the horizontal and vertical components of the reaction at B .

24. AB and AC are two similar uniform rods each of weight W hinged freely at A . Ends B and C are attached to a light inelastic string and are at rest, placed symmetrically on two smooth planes inclined at an angle α to the horizontal each. BC is placed horizontally and A is located above BC . ABC is in a vertical plane. Show that the reaction at B is of magnitude $W \sec \alpha$.

Show that the tension in the string is $\frac{W}{2}(\tan \theta - 2 \tan \alpha)$ where angle $BAC = 2\theta$, with $\tan \theta > 2 \tan \alpha$.

Show also that the reaction of A is a horizontal force of magnitude $\frac{W}{2} \tan \theta$.

25. Two rods AB, BC of weight w and length $2l$ are connected smoothly at B and kept on a smooth cylinder whose radius r and axis horizontal. The rods are symmetrically in equilibrium in a vertical plane by joining the ends A and C from a light inextensible string, which does not touch the cylinder.

If $ABC = 2\theta$ show that the tension in the string $\frac{w}{2l}(r \operatorname{cosec}^2 \theta - l \tan \theta)$.

26. Two smooth uniform rods AB, BC each of length $2a$ and weight W , are freely hinged at B , and are suspended by two light inextensible strings AO, CO each of length $2a$, tied to a fixed point O . A uniform sphere of weight W and radius $\frac{a}{3}$ rests in contact with the rods and is supported by them.

Show that, in the position of equilibrium, each rod makes with the vertical an angle θ given by,

$$\cot^3 \theta + \cot \theta - 30 = 0.$$

Find the only possible value of $\cot \theta$, and hence show that the reaction at the hinge B is W .

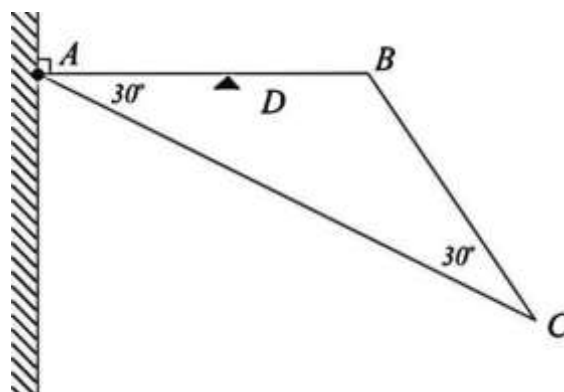
27. AB, BC and CD are uniform light rods of length $l, 2a$ and l respectively and are hinged at the ends B and C . Particles of weight of W are hanging from ends A and D of the rods AB and DC . The system is kept symmetrically in equilibrium in a vertical plane on a fixed smooth cylinder of radius $a\sqrt{3}$ and axis horizontal. If the rod BC is horizontal and in contact with the upper most point of the cylinder, show that the force exerted on the rod AB or CD by the cylinder is of magnitude $\frac{Wl}{2a}$.

Hence show that the reaction between rod BC and the cylinder is $\frac{W}{2a}(4a - l)$.

Find also the horizontal and vertical components of the reaction at joint B .

28. BA and AC are two similar uniform rods of length $2a$ and weight W , which are freely jointed at A . Uniform rod BC of weight w is connected to make a triangular frame work ABC and then the joint B is attached to a smooth hinge. The system is kept in equilibrium so that the rod BC is horizontal with angle $BAC = 2\alpha$ and vertex A below the level of BC , by applying a couple force of moment G on the rod AB . Find the vertical and horizontal components of the forces at joints A and C , in terms of W and w . Show that magnitude of G is $2a(2W + w) \sin \alpha$ to anticlockwise sense.

29. Ends of the uniform rods AB, BC, CA each of weight W are smoothly jointed at ends to form a triangular frame work ABC with $BAC = BCA = 30^\circ$. Frame work is connected to a hinge at A , as shown by the diagram below, and kept so that the rod AB is horizontal by means of a smooth peg touching the rod AB at D where $AD = a$ and $AB = 4a$. Find the horizontal and vertical component of reaction at B and C .



30. $ABCDE$ pentagon is formed of identical uniform rods of weight W each. All hinges are smooth joints. The pentagon is hanged in a vertically plane by means of two smooth light rings at B and E , which are movable on a horizontal wire. The rods AB and AE inclined at an angle θ to the horizontal and rods BC and ED are inclined at an angle ϕ to the horizontal so that, rods BC, CD, DE lie below A . The horizontal distance between B and E exceeds CD . The system is kept as above, by applying two equal and opposite forces at B and E respectively. Show that the magnitude of each of these forces is $\frac{W}{2}(\cot \theta - 2 \cot \phi)$.
31. AB, BC and CD are uniform rods each of length $2a$ and weight W are freely hinged at the points B and C . The rods AB and CD are supported by smooth stationary pegs P and Q so that the rod BC is horizontal and lies above the ends A and D . The pegs are distanced by a length of $2c$ ($PQ = 2c$). Show that, when the system is in equilibrium as mentioned above, reaction at the pegs is of magnitude $\frac{3W}{2 \sin \alpha}$ where, α is the inclination of the rod AB to the vertical. Show also that α given by the relation $\sin^3 \alpha = \frac{3(c-a)}{2a}$. Obtain the reaction at B is as $\frac{W}{2} \sqrt{1 + 9 \cot^2 \alpha}$ with inclination to the horizontal $\tan^{-1} \left(\frac{1}{3} \tan \alpha \right)$.
32. Uniform rods AB, BC, CD of weight w and length $2l$ are smoothly connected together at ends B and C . System is kept in equilibrium on two smooth pegs which are in same horizontal level, by touching the rods AB and CD , so that the rod BC is horizontal. If the rods AB and CD are inclined at an angle α to the vertical show that the distance between the pegs is given by $\frac{2l(3+2 \sin^3 \alpha)}{3}$. Find the vertical and horizontal component of the reaction at B . If the reaction at B is inclined at an angle β to the vertical show that, $\tan \alpha \tan \beta = 3$.
33. Four identical rods AB, BC, CD and AD of weight w and length $2a$ are connected at each ends to form a rhombus $ABCD$. The system is kept so that the vertex C is at vertically above to vertex A and the framework is being supported by two smooth pegs touching the rods AD and AB which are in same horizontal level and distance b apart. Shape of a rhombus with $BAD = 2\alpha$, is given to the framework by a light rod BD . Show that the tension in the string is $\frac{w(b \operatorname{cosec}^2 \alpha - 4a \sin \alpha)}{2a \cos \alpha}$.
34. AB, AC are two uniform identical rods whose length a which are smoothly jointed at end A . BD is a light rod whose length b and is smoothly hinged at B . The end D of the rod BD is connected to a smooth light ring which can move on rod AC . The system is hanged at A over a smooth peg. Show that the inclination of the rod AC to the vertical is $\tan^{-1} \left(\frac{b}{a + \sqrt{a^2 - b^2}} \right)$. Find the stress in rod BD .

35. AB and BC are two uniform rods, of equal length $2a$ and of weight W and $2W$ respectively.

They are smoothly hinged together at B and also hinged at A and C to a fixed horizontal beam.

The rods are in equilibrium in a vertical plane with B below AC and $CAB = \alpha$.

Show that the horizontal component of the reaction of the hinge at B is $\frac{3}{4}W \cot \alpha$.

Find also the vertical component of this reaction. If, further, the lines of action of the reactions at A and C are perpendicular to each other, show that $\tan \alpha = \frac{3}{\sqrt{35}}$.



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